STAFF REVIEW · H446 145 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. Map names such as example.org to relevant resource records including IP addresses.

2. Holding a CPU's working data/ALU results.

3. Binary search: O(log n) rather than O(n).

4. AND.

I Do — source worked steps

1. Send packets 1,2,3 — Identify the relevant information and make a prediction.

2. Receive 2,1,3 — Explain the change using correct technical terms.

3. Reorder 1,2,3 — Packets can follow different routes or encounter different queue delays. With a reliable transport such as TCP, sequence numbers permit correct reordering before delivery of the byte stream; missing data can be retransmitted.

We Do — guided CHECK

1. Reserve dedicated circuit
Identify the relevant information and make a prediction.
Hold resources during call
Explain the change using correct technical terms.
Compare shared packet links
A circuit-switched system sets aside end-to-end resources while active; ordinary packet-switched IP shares links dynamically with others. Circuit switching offers predictable reserved capacity but can waste it during silence; packet switching adapts.

You Do — source mark schemes

1. Packets individually forwarded/shared links [3], circuit pre-establishes reserved route/capacity [3].

2. Different routes [2], variable congestion/queuing [2], sequence numbers and reliable transport reorder [2].

3. Shared capacity efficiency [2], delay/jitter concerns [2], circuit reservation benefits/costs [2], qualified judgement incl transport/QoS [2].

Exit Ticket — source answer

1. False. IP offers best-effort delivery; transport protocols may provide ordering and retransmission.